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main.c

```c
#include <stdio.h>

#define MAX 10
#define INF 9999

int main()
{
    int n;
    int cost[MAX][MAX];
    int dist[MAX][MAX];
    int nextHop[MAX][MAX];
    int i, j, k;
    int updated;

    printf("Enter the number of routers: ");
    scanf("%d", &n);

    printf("\nEnter the cost matrix:\n");
    printf("(Enter 9999 for no direct connection)\n\n");

    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
        {
            scanf("%d", &cost[i][j]);

            dist[i][j] = cost[i][j];

            if (i == j)
                nextHop[i][j] = i;
            else if (cost[i][j] != INF)
                nextHop[i][j] = j;
            else
                nextHop[i][j] = -1;
        }
    }

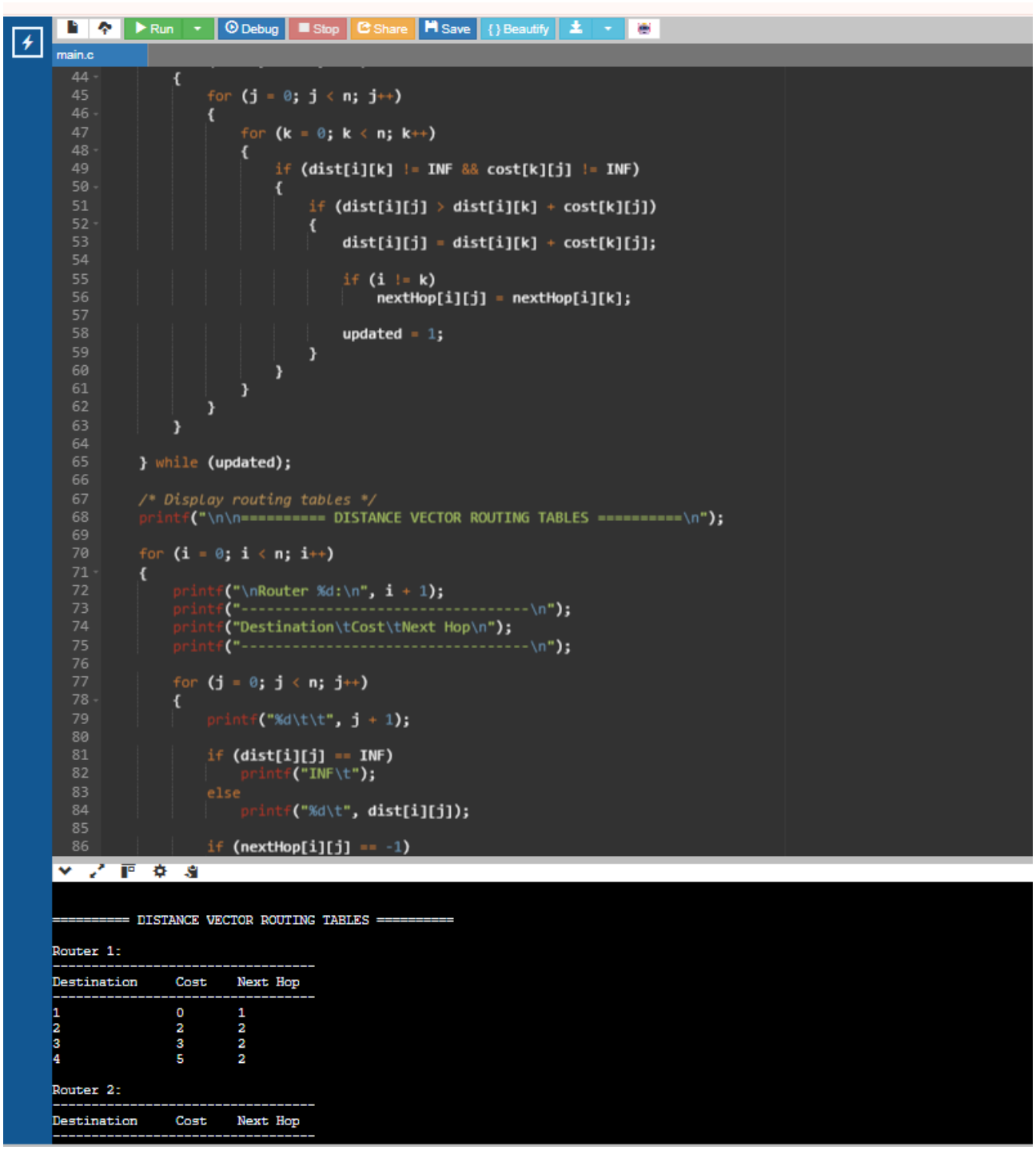
    /* Distance Vector Algorithm */
    do
    {
        updated = 0;

        for (i = 0; i < n; i++)
```

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